

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is: A SYN flood attack

The logs show that: A large number of SYN packets are being sent from the same IP address, **203.0.113.0**, to the web server at **192.0.2.1** on port **443**. The attacker continues sending SYN requests at a rapid rate

This event could be: A direct Denial-of-Service (DoS) attack, specifically a SYN flood attack

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. The client sends a **SYN** packet to request a TCP connection with the web server
2. The web server responds with a **SYN/ACK** packet to accept the connection request
3. The client sends an **ACK** packet to confirm the connection. The TCP connection is then established

Explain what happens when a malicious actor sends a large number of SYN packets all at once: The web server attempts to respond to each SYN request and uses system resources to handle the connection attempts. When the number of requests becomes too large, the server can become overwhelmed and have fewer resources available for legitimate users.

Explain what the logs indicate and how that affects the server: The logs show that the attacker at **203.0.113.0** repeatedly sends SYN packets to the web server at **192.0.2.1**. The server initially responds normally, but as the attack continues, legitimate users begin receiving **RST**, **ACK** packets and **504 Gateway Time-out** errors. Eventually, the server stops responding to legitimate visitor traffic, preventing users from accessing the website